Formula Derivation of the RIEKF
Right-Invariant Extended Kalman Filter

Invariant Error and the Group-Affine Property

$$(\xi^\wedge) : \mathbb{R}^{\dim g} \to g \qquad \text{wedge} \quad \text{axis-angle} \to so3$$

$$\exp(\xi) = \exp_m(\xi^\wedge) \qquad \exp_m \text{ is the general matrix exponential}$$

dynamic: $X_t \in G$

$$\frac{d}{dt}X_t = f_{u_t}(X_t), \qquad \hat{X}_t \text{ is the estimate}$$

$$\text{Right-invariant error:} \quad \eta_t^r = (\hat{X}_t L)^{-1}(X_t L)^{-1} = \hat{X}_t^{-1} X_t^{-1}$$

$$\text{Left-invariant error:} \quad \eta_t^l = (L\hat{X}_t)^{-1}(LX_t) = \hat{X}_t^{-1}\hat{X}_t$$

$\star$ Right multiplication: fixed base, moving body, RIEKF

Group affine

$$f_{u_t}(X_1 X_2) = f_{u_t}(X_1)X_2 + X_1 f_{u_t}(X_2) - X_1 f_{u_t}(I_d)X_2$$

$$\Downarrow$$

$$\frac{d}{dt}\eta_t^r = g_{u_t}(\eta_t^r), \qquad g_{u_t}(\eta^r) = f_{u_t}(\eta^r) - \eta^r f_{u_t}(I_d)$$

$\star$ Right multiplication is convenient: Group affine $\to$ makes the error dynamics non-autonomous

Because:

1. the bias is not part of the group structure
2. discrete jumps

After applying the inverse wedge to obtain autonomy

$$\frac{d}{dt}\eta_t^r = g_{u_t}\big(\exp(\epsilon_t)\big) = (A_t\epsilon_t)^\wedge + O(\|\epsilon\|^2) \qquad \text{first-order Taylor expansion}$$

$$\Downarrow$$

$$\frac{d}{dt}\epsilon_t = A_t\epsilon_t \qquad \text{This is exact because the error system is already linear}$$

Observation model

$$Y_t = X_t^{-1} b + V_t$$

The observation Y is the value of b expressed in another coordinate frame
$\rightarrow$ The error computation compares Y and b in the same coordinate frame.
Residual:

$$r = \hat{X} Y - b$$

When $Y = X^{-1} b$

$$r = \hat{X} X^{-1} b - b$$

$$= \eta^r b - b \rightarrow \text{depends only on the right-invariant error}$$

Adjoint Operator Ad

$$\mathrm{Ad}_X(\hat{\xi}) = X \hat{\xi} X^{-1}$$

After the transformation by Ad_X, $\hat{\xi}$ becomes
the representation in the new coordinate frame after applying transformation X to the original frame
for example, base $\xrightarrow{X}$ body

How to use angular-velocity observations?

The angular-velocity observation is

$$\tilde{\omega}_t = \mathrm{vec}(\tilde{\omega}_x, \tilde{\omega}_y, \tilde{\omega}_z) \in \mathbb{R}^3$$

The Euler angles are $q_t = \mathrm{vec}(q_x, q_y, q_z)$, i.e., the Euler-angle vector

$$q_x = \phi, \qquad q_y = \theta, \qquad q_z = \psi$$

$\rightarrow$ The Euler-angle derivatives are defined in the final coordinate frame $[e_x, e_y, e_z]$,
but the current body angular velocity must be expressed according to the actual rotations
$\rightarrow$ first z, then Y, and then x,

$$R = R_z(\psi)\, R_y(\theta)\, R_x(\phi)$$

$$\omega = \dot{\phi}\, e_x + \dot{\theta}\, R_x(\phi)^T e_y + \dot{\psi} \left(R_y(\theta) R_x(\phi) \right)^T e_z$$

$$e'_x = e_x, \qquad e'_y = R_x(\phi)^T e_y$$

e_y denotes the y-axis of the final coordinate frame, while e'_y denotes the y-axis right after the first rotation
$\rightarrow$ e'_x rotates last, so its rotation axis coincides with the physical axis.
After e'_y rotates, e'_x rotates afterward, so the angle θ defined in the e_y frame already contains the effect of the $R_x(\phi)$ rotation (a rotation about the x-axis also rotates the

y-axis through $R_x(\phi)$). Therefore, we right-multiply by $R_x(\phi)^T = R_x^{-1}(\phi)$ to transform it back.

$$R_x(\phi)^T e_y = \begin{bmatrix} 0 \\ \cos\phi \\ -\sin\phi \end{bmatrix}, \qquad \big(R_y(\theta)R_x(\phi)\big)^T e_z = \begin{bmatrix} -\sin\theta \\ \sin\phi\cos\theta \\ \cos\phi\cos\theta \end{bmatrix}$$

$$\omega = \underbrace{\begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}}_{E(\phi,\theta)} \begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix}$$

$$\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = E(\phi,\theta)^{-1} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

Error Dynamics

Let

$$\delta q_t = q_t - \bar{q}_t \in \mathbb{R}^3$$

be the Euler-angle error between the true and estimated states
$\rightarrow$ Error dynamics:

$$\frac{d}{dt}\delta q_t = g(\delta q_t, \tilde{\omega}_t, q_t) \qquad \rightarrow \text{nonlinear}$$

$$\frac{d}{dt}q_t = \frac{d}{dt}(\bar{q}_t + \delta q_t) \approx f(\bar{q}_t, \omega_t) + \left.\frac{\partial f}{\partial q}\right|_{q=\bar{q}_t} \delta q_t$$

$$= \frac{d}{dt}\bar{q}_t + \frac{d}{dt}\delta q_t \qquad \text{first-order Taylor expansion}$$

The last equality above holds because

$$\frac{d}{dt}\bar{q}_t = f(\bar{q}_t, \omega_t),$$

Therefore

$$\frac{d}{dt}\delta q_t \approx A(\tilde{\omega}_t, \bar{q}_t)\,\delta q_t$$

where A is the Jacobian of f, given by

$$\dot{q}_t = f(q_t, \omega_t) = \begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \begin{bmatrix} \omega_x + (\omega_y \sin\phi + \omega_z \cos\phi)\tan\theta \\ \omega_y \cos\phi - \omega_z \sin\phi \\ \dfrac{\omega_y \sin\phi + \omega_z \cos\phi}{\cos\theta} \end{bmatrix}$$

$$A = \frac{\partial f}{\partial q_t} = \begin{bmatrix} \frac{\partial\dot\phi}{\partial\phi} & \frac{\partial\dot\phi}{\partial\theta} & \frac{\partial\dot\phi}{\partial\psi} \\ \frac{\partial\dot\theta}{\partial\phi} & \frac{\partial\dot\theta}{\partial\theta} & \frac{\partial\dot\theta}{\partial\psi} \\ \frac{\partial\dot\psi}{\partial\phi} & \frac{\partial\dot\psi}{\partial\theta} & \frac{\partial\dot\psi}{\partial\psi} \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{\omega_z \bar{c}_x + \omega_y \bar{s}_x}{\bar{c}_y^2} & \bar{t}_y(\omega_y \bar{c}_x - \omega_z \bar{s}_x) & 0 \\ -(\omega_z \bar{c}_x + \omega_y \bar{s}_x) & 0 & 0 \\ \dfrac{\omega_y \bar{c}_x - \omega_z \bar{s}_x}{\bar{c}_y} & \dfrac{\bar{s}_y(\omega_z \bar{c}_x + \omega_y \bar{s}_x)}{\bar{c}_y^2} & 0 \end{bmatrix}$$

Here $\bar{c}_x = \cos\bar{\phi}, \bar{c}_y = \cos\bar{\theta}, \bar{c}_z = \cos\bar{\phi}$.

Rotation-Matrix Dynamics

Reconstruction equation: the derivative of the rotation is obtained from the angular velocity $\tilde{\omega}_t$ measured in the body frame and the current attitude R_t.

$$R_t \in SO(3), \qquad \frac{d}{dt} R_t = R_t(\tilde{\omega}_t)_\times$$

In group theory, $(\tilde{\omega}_t)_\times \in so(3)$ denotes the derivative of the trajectory at the identity.

$\Downarrow$ we can derive the derivative of the right-invariant error

$$\eta_t = \bar{R}_t R_t^T \in SO(3)$$

$$\dot{R}_t = R_t(\tilde{\omega}_t)_\times, \qquad \dot{\bar{R}}_t = \bar{R}_t(\tilde{\omega}_t)_\times$$

The dynamics are:

$$\frac{d}{dt}\eta_t = \dot{\bar{R}}_t R_t^T + \bar{R}_t \dot{R}_t^T$$

$$= \bar{R}_t(\tilde{\omega}_t)_\times R_t^T + \bar{R}_t\big(R_t(\tilde{\omega}_t)_\times\big)^T$$

Skew-symmetry property:

$$\big((\tilde{\omega}_t)_\times\big)^T = -(\tilde{\omega}_t)_\times$$

$$= \bar{R}_t(\tilde{\omega}_t)_\times R_t^T + \bar{R}_t\big(-(\tilde{\omega}_t)_\times R_t^T\big)$$

$$= \bar{R}_t(\tilde{\omega}_t)_\times R_t^T - \bar{R}_t(\tilde{\omega}_t)_\times R_t^T = 0$$

Because

$$\eta_t = \exp\big((\xi_t)_\times\big)$$

Therefore

$$\frac{d}{dt}\exp\big((\xi_t)_\times\big) = 0$$

$$\frac{d}{dt}\big(I + (\xi_t)_\times\big) \approx 0$$

$$\frac{d}{dt}(\xi_t)_\times = 0 \qquad \rightarrow \qquad \frac{d}{dt}\xi_t = 0$$

$\star$ if and only if R_t and $\bar{R}_t$ are driven by the same angular velocity does it become zero

Only the right-invariant error is zero; the left-invariant error is not.

Right-Invariant EKF — $SE_{N+2}(3)$

$SE_{N+2}(3)$ denotes
1. base rotation: $R_{WB}(t)$
2. velocity: $v_B(t)$,
3. position: $p_{WB}(t)$,
the observed positions of the N feet: from $p_{WC_1}(t)$ to $p_{WC_N}(t)$

$$X_t \triangleq \begin{bmatrix} {}_W R_{WB}(t) & {}_W v_B(t) & {}_W p_{WB}(t) & {}_W p_{WC_1}(t) & \cdots & {}_W p_{WC_N}(t) \\ 0_{1,3} & 1 & 0 & 0 & \cdots & 0 \\ 0_{1,3} & 0 & 1 & 0 & \cdots & 0 \\ 0_{1,3} & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_{1,3} & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

The lower-left block represents the Euclidean coordinates.

For each contact point, the process and measurement models are the same
$\rightarrow$ assume there is only one contact point.
State:

$$X_t = \begin{bmatrix} R_t & v_t & p_t & d_t \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix}$$

Input

$$u_t = \begin{bmatrix} {}_B\tilde{\omega}_{WB}(t) \\ {}_B\tilde{a}_{WB}(t) \end{bmatrix} = \begin{bmatrix} \tilde{\omega}_t \\ \tilde{a}_t \end{bmatrix}$$

that is, the angular velocity and acceleration in the body frame

Lie Algebra and Adjoint of $SE_{N+2}(3)$

$$For$$

SE$_{N+2}(3)$, $(\cdot)^\wedge : \mathbb{R}^{3N+9} \rightarrow \mathfrak{g}$

$$3N: \ 3Dpositionsofthe$$

N$feet$

$$3+3+3: \text{ base axis-angle} + \text{velocity} + \text{translation (from time 0 to } t)$$

$\star$ This state matrix does not include joint rotations, because they do not need to be estimated

$$\rightarrow \text{ wedge matrix}(\cdot)^\wedge \text{ is:}$$

$$\xi^\wedge = \begin{bmatrix} (\xi^R)_\times & \xi^v & \xi^p & \xi^d \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix} \in \mathfrak{g}$$

The adjoint is

$$\mathrm{Ad}_X(\xi) = \left(X\xi^\wedge X^{-1}\right)^\vee$$

$$\left(X\xi^\wedge X^{-1}\right) = \begin{bmatrix} R & v & p & d \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} (\omega)_\times & a & b & c \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} R^T & -R^Tv & -R^Tp & -R^Td \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (R\omega)_\times & Ra + (v)_\times R\omega & Rb + (p)_\times R\omega & Rc + (d)_\times R\omega \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} R^T & 0 & 0 & 0 \\ (v)_\times R^T & R^T & 0 & 0 \\ (p)_\times R^T & 0 & R^T & 0 \\ (d)_\times R^T & 0 & 0 & R^T \end{bmatrix}}_{\mathrm{Ad}_X} \begin{bmatrix} \omega \\ a \\ b \\ c \end{bmatrix}$$

this is the coordinate transformation from the world frame to the body frame at time t.

Continuous-Time IMU Dynamics

The IMU can be modeled as

$$\tilde{\omega}_t = \omega_t + w_t^g, \qquad \tilde{a}_t = a_t + w_t^a$$

The second term is Gaussian white noise, with zero covariance across different time instants

When a 3D sensor detects contact, the contact point is assumed to be fixed in the world frame:

$$\rightarrow \text{the measured velocity in the world frame is zero}$$

$$_W\tilde{v}_C = 0_{3,1} = {}_c v_C + w_t^v$$

Here the subscript c denotes contact, and the superscript/prefix c denotes the true velocity

$$\Downarrow$$

System Dynamics

$$\begin{cases} \dfrac{d}{dt}R_t = R_t(\tilde{\omega}_t - w_t^g)_\times = R_t(\tilde{\omega}_t)_\times - R_t(w_t^g)_\times \\ \dfrac{d}{dt}V_t = R_t(\tilde{a}_t - w_t^a) + g \\ \dfrac{d}{dt}P_t = V_t \\ \dfrac{d}{dt}d_t = R_t h_R(\tilde{\alpha}_t)(-w_t^v) \end{cases}$$

In the first two equations, R_t plays the same role: it maps the angular velocity/acceleration measured by the IMU from the body frame to
the world frame

$$(2) \quad \tilde{a}_t = R_t^T(\dot{V}_t - g) + w_t^a \qquad \text{IMU data}$$

$\tilde{a}_t$ denotes the acceleration measured by the IMU, while $\dot{V}_t$ denotes the true acceleration

$$\star \text{ specific force:} \quad f_{\text{spec}} = \frac{F_{\text{non-gravitational}}}{m}, \qquad \text{force measured by the IMU}$$

$$\frac{d}{dt}d_t = R_t\, h_R(\tilde{\alpha}_t)(-w_t^v)\textbf{How should this be interpreted?}$$

$h_R(\tilde{\alpha}_t)$ is the rotation matrix computed from the joint-encoder reading $\tilde{\alpha}_t$
computed rotation matrix

$$\text{Step1: Foot frame} \xrightarrow{h_R(\tilde{\alpha}_t)} \text{Base frame (IMU)}$$

$$\text{Step2: Base frame} \xrightarrow{R_t} \text{World frame}$$

The velocity of the foot itself in the foot frame is pure noise

$$\rightarrow \; -w_t^v$$

$\Downarrow$ Matrix form of the system dynamics: this is constructed directly from the equations

$$\frac{d}{dt}X_t = \begin{bmatrix} R_t(\tilde{\omega}_t)_\times & R_t\tilde{a}_t + g & v_t & 0_{3,1} \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} R_t & v_t & p_t & d_t \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} (w_t^g)_\times & w_t^a & 0_{3,1} & h_R(\tilde{\alpha}_t)w_t^v \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix}$$

$$= f_{u_t}(X_t) - X_t w_t^\wedge \qquad \text{Body-frame noise}$$

$\star$ satisfies the Group-affine property

For the right-invariant error

$$\eta_t^r = \bar{X}_t X_t^{-1} \qquad L^{-1}L = I$$

$$f_{u_t}(\eta_t^r X_t) = f_{u_t}(\eta_t^r)X_t + \eta_t^r f_{u_t}(X_t) - \eta_t^r f_{u_t}(I_d)X_t$$

Autonomous Error Dynamics

$$\frac{d}{dt}\eta_t^r = \frac{d}{dt}\big(\bar{X}_t X_t^{-1}\big) = \dot{\bar{X}}_t X_t^{-1} - \bar{X}_t(X_t^{-1}\dot{X}_t X_t^{-1})$$

$$\begin{cases} \dot{\bar{X}} = f_{u_t}(\bar{X}_t) & \rightarrow \text{because the dynamics of the estimate } \bar{X} \text{ are noise-free} \\ \dot{X} = f_{u_t}(X_t) - X_t w_t^\wedge \end{cases}$$

$$\Downarrow$$

$$\frac{d}{dt}\eta_t^r = f_{u_t}(\bar{X}_t)X_t^{-1} - \bar{X}_t X_t^{-1}\big(f_{u_t}(X_t) - X_t w_t^\wedge\big)X_t^{-1}$$

$$= f_{u_t}(\eta_t^r) + \eta_t^r f_{u_t}(X_t)X_t^{-1} - \eta_t^r f_{u_t}(I_d) - \eta_t^r\big(f_{u_t}(X_t) - X_t w_t^\wedge\big)X_t^{-1}$$

$$= f_{u_t}(\eta_t^r) - \eta_t^r f_{u_t}(I_d) + \bar{X}_t w_t^\wedge X_t^{-1}$$

$$= f_{u_t}(\eta_t^r) - \eta_t^r f_{u_t}(I_d) + \big(\bar{X}_t w_t^\wedge \bar{X}_t^{-1}\big)\eta_t^r$$

$$= g_{u_t}(\eta_t^r) + \bar{w}_t^\wedge \eta_t^r \qquad \bar{w}_t^\wedge := \mathrm{Ad}_{\bar{X}_t} w_t^\wedge$$

$\star$ this is the autonomous error dynamics

The form of g comes from Theory 1 (obtained from invariant error + group affine):

$$g_{u_t}(\eta^r) = f_{u_t}(\eta^r) - \eta^r f_{u_t}(I_d)$$

$$\Downarrow \text{ further logarithmic linearization}$$

$$\frac{d}{dt}\eta_t^r = g_{u_t}(\eta_t^r) + \bar{w}_t^\wedge \eta_t^r$$

$$\begin{cases} g_{u_t}(I+\xi_t^\wedge) \approx g_{u_t}(I) + \left.\dfrac{\partial g_{u_t}}{\partial \eta}\right|_{\eta=I} \xi_t^\wedge \\ \qquad\qquad = g_{u_t}(I) + \left(\left.\dfrac{\partial\, g_{u_t}(I+\xi^\wedge)}{\partial \xi}\right|_{\xi=0} \xi_t\right)^\wedge = (A_t^r \xi_t)^\wedge \\ \bar{w}_t^\wedge(I+\xi_t^\wedge) = \bar{w}_t^\wedge + \bar{w}_t^\wedge \xi_t^\wedge \approx \bar{w}_t^\wedge \qquad \text{neglecting second-order infinitesimals} \end{cases}$$

$$\frac{d}{dt}\eta_t^r \approx (A_t^r \xi_t)^\wedge + \bar{w}_t^\wedge$$

$$\frac{d}{dt}\xi_t = A_t^r \xi_t + \bar{w}_t$$

$$= A_t^r \xi_t + \mathrm{Ad}_{\bar{X}_t} w_t$$

$$\begin{cases} g_{u_t}(I) = 0 \\ \bar{w}_t^\wedge \xi_t^\wedge \approx 0 \\ \quad A_t^r = \left.\dfrac{\partial g_{u_t}}{\partial \xi}\right|_{\xi=0} \qquad \text{the explicit form of } A \text{ will be defined next} \end{cases}$$

$$\frac{d}{dt}\xi_t = A_t^r \xi_t + \mathrm{Ad}_{\bar{X}_t} w_t \textbf{the form of } A \textbf{ in}$$

$$g_{u_t}\big(\exp(\xi_t)\big) \approx \underbrace{\begin{bmatrix} \big(I + (\xi_t^R)_\times\big)(\tilde{\omega}_t)_\times & \big(I + (\xi_t^R)_\times\big)\tilde{a}_t + g & \xi_t^v & 0_{3,1} \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix}}_{f_{u_t}(\exp(\xi_t))}$$

$$- \underbrace{\begin{bmatrix} I + (\xi_t^R)_\times & \xi_t^v & \xi_t^p & \xi_t^d \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} (\tilde{\omega}_t)_\times & \tilde{a}_t + g & 0_{3,1} & 0_{3,1} \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix}}_{\exp(\xi_t^\wedge)\, f_{u_t}(I_d) \text{ is the first-order approximation; in the left term, } I+\xi^\wedge \text{ is the first-order approximation of the exponential map}}$$

$$= \begin{bmatrix} 0_{3,3} & (g)_\times \xi_t^R & \xi_t^v & 0_{3,1} \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0_{3,1} \\ (g)_\times \xi_t^R \\ \xi_t^v \\ 0_{3,1} \end{bmatrix}^\wedge$$

$$\frac{d}{dt}\eta_t^r = g_{u_t}(\eta_t^r) + \bar{w}_t^\wedge \eta_t^r, \qquad g_{u_t}(\exp(\xi_t)) \approx (A\xi_t)^\wedge$$

$$\therefore \quad A\xi_t = \begin{bmatrix} 0_{3,1} \\ (g)_\times \xi_t^R \\ \xi_t^v \\ 0_{3,1} \end{bmatrix}$$

$$\text{Hence} \quad A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ (g)_\times & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Motion Model and Right-Invariant Propagated Observation Model

$$\frac{d}{dt}\bar{X}_t = f_{u_t}(\bar{X}_t), \qquad \frac{d}{dt}P_t = A_t P_t + P_t A_t^T + \bar{Q}_t$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ (g)_\times & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \text{and} \qquad \bar{Q} = \mathrm{Ad}_{\bar{X}_t} \mathrm{Cov}(w_t) \mathrm{Ad}_{\bar{X}_t}^T$$

This shows the core purpose of deriving autonomous error dynamics: to convert the nonlinear error system on the group into a linear stochastic system in the Lie algebra

$$\dot{\xi}_t = A_t \xi_t + \text{noise term.}$$

Once this form is obtained, the covariance can be propagated as a continuous-time linear system (this is the Riccati principle):

$$\dot{P}_t = A_t P_t + P_t A_t^T + \bar{Q}_t.$$

Therefore, the previous derivation of autonomous error dynamics and logarithmic linearization is ultimately for obtaining A.

Observation model

$\alpha_t \in \mathbb{R}^N$ denotes joint coordinates or rotations obtained from encoders, and is not part of the estima

$\rightarrow$ The coordinate transformation between the rigid body and the contact point is also affected by noise

$$\tilde{\alpha}_t = \alpha_t + w_t^\alpha, \qquad w_t^\alpha \sim \mathcal{N}(0_{N,1}, \Sigma^\alpha)$$

Position of the contact point relative to the base

$$_B p_{BC}(t) = h_p(\tilde{\alpha}_t - w_t^\alpha) \approx h_p(\tilde{\alpha}_t) - J_p(\tilde{\alpha}_t) w_t^\alpha$$

$h_R(\tilde{\alpha}_t)$ is computed from the joint-encoder reading $\tilde{\alpha}_t$the position vector of the contact point in the l

$$\Downarrow$$

$$h_p(\tilde{\alpha}_t) = R_t^T(d_t - p_t) + J_p(\tilde{\alpha}_t) w_t^\alpha$$

The vector from the base to the contact point is defined in the world frame, so it must first be transfo

$$h_p(\tilde{\alpha}_t) = R_t^T(d_t - p_t) + J_p(\tilde{\alpha}_t) w_t^\alpha \textbf{in matrix form}$$

$$\underbrace{\begin{bmatrix} h_p(\tilde{\alpha}_t) \\ 0 \\ 1 \\ -1 \end{bmatrix}}_{Y_t} = \underbrace{\begin{bmatrix} R^T & -R^T v & -R^T p & -R^T d \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix}}_{X_t^{-1}} \underbrace{\begin{bmatrix} 0_{3,1} \\ 0 \\ 1 \\ -1 \end{bmatrix}}_{b} + \underbrace{\begin{bmatrix} J_p(\tilde{\alpha}_t) w_t^\alpha \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{V_t}$$

This naturally constructs the structure of the right-invariant observation model, where J_p is the Jacob

$$Y_t = X_t^{-1} b + V_t$$

The reference point b is in the world frame; after X_t^{-1} maps the reference point b into the rigid-body frame, adding observation noise yields the measurement Y_t.

$\rightarrow$ innovation: multiplicative update (using the observation)

$$\bar{X}_t^+ = \exp\big(L_t(\bar{X}_t Y_t - b)\big)\, \bar{X}_t$$

$$\begin{cases} \eta_t^{r+} = \bar{X}_t^+ X_t^{-1} = \exp\big(L_t(\bar{X}_t Y_t - b)\big) \bar{X}_t X_t^{-1} \\ \qquad = \exp\big(L_t(\eta_t^r b - b + \bar{X}_t V_t)\big) \eta_t^r \end{cases}$$

This is because

$$\bar{X}_t Y_t = \bar{X}_t(X_t^{-1} b + V_t) = \eta_t^r b + \bar{X}_t V_t$$

If $\bar{X}_t = X_t$ is estimated correctly and the noise $V_t = 0$, then

$$\bar{X}_t Y_t = b,$$

Hence the residual is

$$\bar{X}_t Y_t - b = R \cdot h_p(\tilde{\alpha}) + p - d$$

Updated state:

$$\bar{X}_t^+ = \exp(\text{gain} \times \text{residual}) \times \text{prior prediction}$$

Posterior error:

$$\eta_t^{r+} = \bar{X}_t^+ X_t^{-1} = \exp\big(L_t(\bar{X}_t Y_t - b)\big)\bar{X}_t X_t^{-1}.$$

Reduced-Dimension Kalman Gain

$\rightarrow$ because in $\bar{X}_t Y_t - b$, the last three rows are all 0

$$\bar{X}_t Y_t - b = \begin{bmatrix} \bar{R}_t & \bar{v}_t & \bar{p}_t & \bar{d}_t \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} h_p(\tilde{\alpha}_t) \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0_{3,1} \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} \bar{R}_t h_p(\tilde{\alpha}_t) + \bar{p}_t - \bar{d}_t \\ 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{6\times 1} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Let

$$L_t(\bar{X}_t Y_t - b) = K_t \Pi\, \bar{X}_t Y_t, \qquad \Pi = \begin{bmatrix} I_3 & 0_{3,3} \end{bmatrix}$$

the original high-dimensional matrix L_t is reduced to K_t, and Π performs the corresponding dimensionality reduction on $\bar{X}_t Y_t$.

$$\Pi\, \bar{X}_t Y_t = \Pi \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \bar{R}_t h_p(\tilde{\alpha}_t) + \bar{p}_t - \bar{d}_t$$

That is, the operator Π keeps only the first 3 dimensions of the 6D vector, so $K_t \in \mathbb{R}^{3\times 3}$, and the b term can be omitted automatically.

$$\begin{cases} \bar{X}_t^+ = \exp\big(K_t \Pi(\bar{X}_t Y_t - b)\big)\bar{X}_t = \exp\big(K_t \Pi \bar{X}_t Y_t\big)\bar{X}_t \\ \eta_t^{r+} = \exp\big(K_t \Pi \bar{X}_t Y_t\big)\bar{X}_t X_t^{-1} = \exp\big(K_t \Pi(\eta_t^r b + \bar{X}_t V_t)\big)\eta_t^r \end{cases}$$

Under first-order Taylor expansion,

$$\eta_t^r \approx I_d + \xi_t^\wedge$$

Thus (again neglecting second-order infinitesimals here)

$$\eta_t^{r+} = \exp\big(K_t\Pi(\eta_t^r b + \bar{X}_t V_t)\big)\eta_t^r \approx \Big(I_d + \big(K_t\Pi(\eta_t^r b + \bar{X}_t V_t)\big)^\wedge\Big)(I_d + \xi_t^\wedge)$$

$$\approx I_d + \xi_t^\wedge + \big(K_t\Pi(\eta_t^r b + \bar{X}_t V_t)\big)^\wedge$$

And because

$$\eta_t^r b \approx (I_d + \xi_t^\wedge)\begin{bmatrix} 0_{3,1} \\ 0 \\ 1 \\ -1 \end{bmatrix}, \qquad \bar{X}_t V_t = \bar{X}_t \begin{bmatrix} J_p(\tilde{\alpha}_t) w_t^\alpha \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence

$$\eta_t^{r+} \approx I_d + \xi_t^\wedge + \left(K_t\Pi\left((I_d + \xi_t^\wedge)\begin{bmatrix} 0_{3,1} \\ 0 \\ 1 \\ -1 \end{bmatrix} + \bar{X}_t \begin{bmatrix} J_p(\tilde{\alpha}_t) w_t^\alpha \\ 0 \\ 0 \\ 0 \end{bmatrix}\right)\right)^\wedge = I_d + \xi_t^{+\wedge}$$

By comparing the expressions above, we obtain the first-order update relation for the posterior error.

$$\text{Hence } \xi_t^{+\wedge} = \xi_t^\wedge + \left(K_t\Pi\left(\begin{bmatrix} I + (\xi_t^R)_\times & \xi_t^v & \xi_t^p & \xi_t^d \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0_{3,1} \\ 0 \\ 1 \\ -1 \end{bmatrix} + \bar{X}_t \begin{bmatrix} J_p(\tilde{\alpha}_t) w_t^\alpha \\ 0 \\ 0 \\ 0 \end{bmatrix}\right)\right)^\wedge$$

$$= \xi_t^\wedge + \left(K_t\Pi\left(\begin{bmatrix} \xi_t^p - \xi_t^d \\ 0 \\ 0 \\ 0 \end{bmatrix} + \bar{X}_t \begin{bmatrix} J_p(\tilde{\alpha}_t) w_t^\alpha \\ 0 \\ 0 \\ 0 \end{bmatrix}\right)\right)^\wedge$$

$$= \xi_t^\wedge + \big(K_t\big([\,0_{3,3} \;\; 0_{3,3} \;\; I \;\; -I\,]\xi_t - \bar{R}_t(J_p(\tilde{\alpha}_t) w_t^\alpha)\big)\big)^\wedge \qquad \downarrow \text{ inverse wedge transform}$$

$$\xi_t^+ = \xi_t + K_t\Big([\,0_{3,3} \;\; 0_{3,3} \;\; I \;\; -I\,]\xi_t + \bar{R}_t(J_p(\tilde{\alpha}_t) w_t^\alpha)\Big)$$

$$= \xi_t - K_t\Big([\,0_{3,3} \;\; 0_{3,3} \;\; -I \;\; I\,]\xi_t - \bar{R}_t(J_p(\tilde{\alpha}_t) w_t^\alpha)\Big)$$

$\xi_t + = \xi_t - K_t\big(H_t\xi_t - \bar{R}_t(J_p(\tilde{\alpha}_t) w_t^\alpha)\big)$this is the observation update equation for the error state

$$\star \;\rightarrow\; \xi - KH\xi + \text{noise} = (I - KH)\xi + \text{noise}$$

Thus we obtain $P_t^+ = (I - K_t H_t)P_t(I - K_t H_t)^T + K_t R K_t^T$ form!

$\star$ **Observation Update Equation!**

$$\begin{cases} \bar{X}_t^+ = \exp(K_t \Pi\, \bar{X}_t Y_t)\, \bar{X}_t \\ P_t^+ = (I - K_t H_t)\, P_t\, (I - K_t H_t)^T + K_t \bar{N}_t K_t^T \\ S_t = H_t P_t H_t^T + \bar{N}_t, \qquad K_t = P_t H_t^T S_t^{-1} \\ H_t = [\, 0_{3,3} \;\; 0_{3,3} \;\; -I \;\; I \,] \\ \bar{N}_t = \bar{R}_t\, J_p(\tilde{\alpha}_t)\, \mathrm{Cov}(w_t^\alpha)\, J_p(\tilde{\alpha}_t)^T\, \bar{R}_t^T \end{cases}$$

The observation matrix $H \rightarrow$ implements the mapping from the error state ξ_t to the observation error

Extension with IMU Bias

After including the bias

$$\tilde{\omega}_t = \omega_t + b_t^g + w_t^g$$

$$\tilde{a}_t = a_t + b_t^a + w_t^a$$

The noise is white and independent across time, and the biases form a parameter vector that is estimated as part of the RIEKF state

$$\theta_t = \begin{bmatrix} b^g(t) \\ b^a(t) \end{bmatrix} = \begin{bmatrix} b_t^g \\ b_t^a \end{bmatrix} \in \mathbb{R}^6$$

The augmented right-invariant error is defined as

$$e_t^r = (\bar{X}_t X_t^{-1},\, \bar{\theta}_t - \theta_t) = (\eta_t^r, \zeta_t)$$

where

$$\eta_t^r = \begin{bmatrix} \bar{R}_t & \bar{v}_t & \bar{p}_t & \bar{d}_t \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R^T & -R^T v & -R^T p & -R^T d \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \bar{R}_t R_t^T & \bar{v}_t - \bar{R}_t R_t^T v_t & \bar{p}_t - \bar{R}_t R_t^T p_t & \bar{d}_t - \bar{R}_t R_t^T d_t \\ 0_{1,3} & 1 & 0 & 0 \\ 0_{1,3} & 0 & 1 & 0 \\ 0_{1,3} & 0 & 0 & 1 \end{bmatrix}$$

$$\zeta_t = \begin{bmatrix} \bar{b}_t^g - b_t^g \\ \bar{b}_t^a - b_t^a \end{bmatrix} = \begin{bmatrix} \zeta_t^g \\ \zeta_t^a \end{bmatrix}$$

System Dynamics

$$\begin{cases} \dfrac{d}{dt}R_t = R_t\big(\tilde{\omega}_t - b_t^g - w_t^g\big)_\times \\ \dfrac{d}{dt}V_t = R_t\big(\tilde{a}_t - b_t^a - w_t^a\big) + g \\ \dfrac{d}{dt}P_t = V_t \\ \dfrac{d}{dt}d_t = R_t\, h_R(\tilde{\alpha}_t)(-w_t^v) \end{cases}$$

IMU bias dynamics $\rightarrow$ pure Brownian motion

$$\begin{cases} \dfrac{d}{dt}b_t^g = w_t^{bg} \\ \dfrac{d}{dt}b_t^a = w_t^{ba} \end{cases} \qquad \text{white noise}$$

$$f_{u_t}(\bar{X}_t, \bar{\theta}_t) = \begin{bmatrix} \bar{R}_t(\bar{\omega}_t)_\times & \bar{R}_t\bar{a}_t + g & \bar{V}_t & 0_{3,1} \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \bar{R}_t(\tilde{\omega}_t - \bar{b}_t^g)_\times & \bar{R}_t(\tilde{a}_t - \bar{b}_t^a) + g & \bar{V}_t & 0_{3,1} \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \\ 0_{1,3} & 0 & 0 & 0 \end{bmatrix}$$

$$e_t^r = (\bar{X}_t X_t^{-1},\, \bar{\theta}_t - \theta_t) = (\eta_t^r, \zeta_t) \quad \text{taking the time derivative}$$

$$\rightarrow \quad \frac{d}{dt}e_t^r = \left(\frac{d}{dt}\eta_t^r,\, \begin{bmatrix} w_t^{bg} \\ w_t^{ba} \end{bmatrix}\right)$$

This is because

$$\frac{d}{dt}\zeta_t = \begin{bmatrix} \dfrac{d}{dt}\big(\bar{b}_t^g - b_t^g\big) \\ \dfrac{d}{dt}\big(\bar{b}_t^a - b_t^a\big) \end{bmatrix} = \begin{bmatrix} 0 - w_t^{bg} \\ 0 - w_t^{ba} \end{bmatrix} = -\begin{bmatrix} w_t^{bg} \\ w_t^{ba} \end{bmatrix}.$$

We can see that the original text missed a minus sign here, so this is an error.

Each Term in the Invariant Error Dynamics

$$① \quad \frac{d}{dt}\big(\bar{R}_t R_t^T\big)$$

$$\frac{d}{dt}\big(\bar{R}R^T\big) = \dot{\bar{R}}R^T + \bar{R}\dot{R}^T$$

$$= \bar{R}(\tilde{\omega} - \bar{b}^g)_\times R^T + \bar{R}\big[-(\tilde{\omega} - b^g - w^g)_\times\big]R^T$$

$$= \bar{R}\big(b^g - \bar{b}^g + w^g\big)_\times R^T$$

$$\text{because } \zeta^g = \bar{b}^g - b^g, \text{ so}$$

$$= \bar{R}(w^g - \zeta^g)_\times R^T$$

$$= \big(\bar{R}(w^g - \zeta^g)\big)_\times$$

$$② \quad \frac{d}{dt}\big(\bar{v}_t - \bar{R}_t R_t^T v_t\big)$$

$$= \dot{\bar{v}}_t - \frac{d}{dt}\big(\bar{R}R^T\big)v_t - \bar{R}R^T\dot{v}_t \qquad \rightarrow \text{sensor data}$$

$$\text{Because} \begin{cases} \dot{v}_t = R_t(\tilde{a}_t - b_t^a - w_t^a) + g, \\ \dot{\bar{v}}_t = \bar{R}_t(\tilde{a}_t - \bar{b}_t^a) + g \end{cases}$$

$$\text{i. Terms related to } g\text{:} \quad g - \bar{R}R^T g = (I - \eta_t^r)g$$

$$\approx \big(I - (I + \xi_t^{R\wedge})\big)g = -\xi_t^{R\wedge} g = (g)_\times \xi_t^R$$

$$\text{ii. Second term:} \quad -\big(\bar{R}(w^g - \zeta^g)\big)_\times v = (v)_\times \bar{R}(w^g - \zeta^g)$$

$$\text{iii. IMU noise term:} \quad \bar{R}_t(\tilde{a}_t - \bar{b}_t^a) - \bar{R}_t R_t^T R_t(\tilde{a}_t - b_t^a - w_t^a)$$

$$= \bar{R}_t(w_t^a - \zeta_t^a)$$

$$\rightarrow \ (g)_\times \xi_t^R + (v)_\times \bar{R}(w^g - \zeta^g) + \bar{R}_t(w_t^a - \zeta_t^a)$$

$$③ \quad \frac{d}{dt}\big(\bar{p}_t - \bar{R}_t R_t^T p_t\big)$$

$$= \dot{\bar{p}}_t - \frac{d}{dt}\big(\bar{R}R^T\big)p_t - \bar{R}R^T\dot{p}_t$$

$$\text{Because} \begin{cases} \dot{\bar{p}}_t = \bar{v}_t, \\ \dot{p}_t = v_t, \end{cases} \qquad \bar{v}_t - \bar{R}R^T v_t = \xi_t^v$$

$$\rightarrow \ \xi_t^v + (p)_\times \bar{R}(w^g - \zeta^g)$$

$$④ \quad \frac{d}{dt}\big(\bar{d}_t - \bar{R}_t R_t^T d_t\big)$$

$$= \dot{\bar{d}}_t - \frac{d}{dt}\big(\bar{R}R^T\big)d_t - \bar{R}R^T\dot{d}_t$$

$$\text{because } \dot{\bar{d}}_t = 0, \qquad \text{the estimator assumes the contact point is stationary}$$

substituting d_t gives

$$\rightarrow \ (d)_\times \bar{R}(w^g - \zeta^g) - \bar{R}R^T\big(R\, h_R(\tilde{\alpha})(-w^v)\big)$$

$$= (d)_\times \bar{R}(w^g - \zeta^g) + \bar{R}\, h_R(\tilde{\alpha})\, w^v$$

Invariant Error Dynamics

$$\begin{cases} \dfrac{d}{dt}\big(\bar{R}_t R_t^T\big) = \big(\bar{R}_t(w^g - \zeta^g)\big)_\times \\ \dfrac{d}{dt}\big(\bar{v}_t - \bar{R}_t R_t^T v_t\big) = (g)_\times \xi_t^R + (v)_\times \bar{R}_t(w^g - \zeta^g) + \bar{R}_t(w_t^a - \zeta_t^a) \\ \dfrac{d}{dt}\big(\bar{p}_t - \bar{R}_t R_t^T p_t\big) = \xi_t^v + (p)_\times \bar{R}_t(w^g - \zeta^g) \\ \dfrac{d}{dt}\big(\bar{d}_t - \bar{R}_t R_t^T d_t\big) = (d)_\times \bar{R}_t(w^g - \zeta^g) + \bar{R}_t h_R(\tilde{\alpha})\, w^v \end{cases}$$

Motion Equation and Observation Equation

Therefore, after augmenting the bias error according to this system of equations, we can construct a linear system

$$\frac{d}{dt}\begin{bmatrix}\xi_t \\ \zeta_t\end{bmatrix} = A_t \begin{bmatrix}\xi_t \\ \zeta_t\end{bmatrix} + \begin{bmatrix}\mathrm{Ad}_{\bar{X}_t} & 0_{12,6} \\ 0_{6,12} & I_6\end{bmatrix} w_t$$

$$\text{where} \ \ w_t = \mathrm{vec}\Bigg(\underbrace{w_t^g,\ w_t^a}_{\text{IMU noise}},\ \underbrace{0_{3,1}}_{p\text{-noise}},\underbrace{h_R(\tilde{\alpha}_t)w_t^v}_{d\text{-noise}},\ \underbrace{w_t^{bg},\ w_t^{ba}}_{\text{bias process noise}}\Bigg)$$

Observation Equation
The equation without bias is

$$\xi_t^+ = \xi_t - K_t\Big(H_t\xi_t - \bar{R}_t\big(J_p(\tilde{\alpha}_t)w_t^\alpha\big)\Big)$$

$$\Downarrow \ \text{here we only augment } \zeta_t$$

$$\begin{bmatrix}\xi_t^+ \\ \zeta_t^+\end{bmatrix} = \begin{bmatrix}\xi_t \\ \zeta_t\end{bmatrix} - \begin{bmatrix}K_t^\xi \\ K_t^\zeta\end{bmatrix}\left(H_t\begin{bmatrix}\xi_t \\ \zeta_t\end{bmatrix} - \bar{R}_t\big(J_p(\tilde{\alpha}_t)w_t^\alpha\big)\right)$$

Final Continuous-Time RIEKF

$$\begin{cases} \dfrac{d}{dt}(\bar{X}_t, \bar{\theta}_t) = \big(f_{u_t}(\bar{X}_t, \bar{\theta}_t),\, 0_{6,1}\big) \\ \dfrac{d}{dt}P_t = A_t P_t + P_t A_t^T + \bar{Q}_t \end{cases}$$

$\rightarrow$ Matrix form of the motion equation

$$\begin{bmatrix} \mathrm{Ad}_{\bar{X}_t} & 0_{12,6} \\ 0_{6,12} & I_6 \end{bmatrix} w_t$$

$$= \begin{bmatrix} \bar{R}_t & 0_{3,3} & 0 & 0 & 0 & 0 \\ (\bar{v}_t)_\times \bar{R}_t & \bar{R}_t & 0 & 0 & 0 & 0 \\ (\bar{p}_t)_\times \bar{R}_t & 0 & \bar{R}_t & 0 & 0 & 0 \\ (\bar{d}_t)_\times \bar{R}_t & 0 & 0 & \bar{R}_t & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{3\times 3} & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{3\times 3} \end{bmatrix} \begin{bmatrix} w_t^g \\ w_t^a \\ 0 \\ h_R(\tilde{\alpha}_t) w_t^v \\ w_t^{bg} \\ w_t^{ba} \end{bmatrix}$$

$$= \begin{bmatrix} \bar{R}_t w_t^g \\ \bar{R}_t w_t^a + (\bar{v}_t)_\times \bar{R}_t w_t^g \\ \bar{R}_t \cdot 0 + (\bar{p}_t)_\times \bar{R}_t w_t^g \\ \bar{R}_t \cdot h_R(\tilde{\alpha}_t) w_t^v + (\bar{d}_t)_\times \bar{R}_t w_t^g \\ w_t^{bg} \\ w_t^{ba} \end{bmatrix}$$

Compare and cancel the terms in the following system of equations

$$\begin{cases} \dfrac{d}{dt}\big(\bar{R}_t R_t^T\big) = \big(\bar{R}_t (w^g - \zeta^g)\big)_\times = \bar{R}_t w_t^g - \bar{R}_t \zeta_t^g \\ \dfrac{d}{dt}\big(\bar{v}_t - \bar{R}_t R_t^T v_t\big) = (g)_\times \xi_t^R + (v)_\times \bar{R}_t (w^g - \zeta^g) + \bar{R}_t (w_t^a - \zeta_t^a) \\ \dfrac{d}{dt}\big(\bar{p}_t - \bar{R}_t R_t^T p_t\big) = \xi_t^v + (p)_\times \bar{R}_t (w^g - \zeta^g) \\ \dfrac{d}{dt}\big(\bar{d}_t - \bar{R}_t R_t^T d_t\big) = (d)_\times \bar{R}_t (w^g - \zeta^g) + \bar{R}_t h_R(\tilde{\alpha})\, w^v \end{cases}$$

Remaining: $\begin{cases} 0 & 0 & 0 & -\bar{R}_t \zeta_t^a \\ (g)_\times \xi^R & 0 & -(v)_\times \bar{R}_t \zeta_t^a & -\bar{R}_t \zeta_t^g \\ 0 & \xi^v & -(p)_\times \bar{R}_t \zeta_t^g & 0 \\ 0 & 0 & -(d)_\times \bar{R}_t \zeta_t^g & 0 \end{cases}$ $\Downarrow$ obtain A_t

$$A_t = \begin{bmatrix} 0 & 0 & 0 & 0 & -\bar{R}_t & 0 \\ (g)_\times & 0 & 0 & 0 & -(\bar{v}_t)_\times \bar{R}_t & -\bar{R}_t \\ 0 & I & 0 & 0 & -(\bar{p}_t)_\times \bar{R}_t & 0 \\ 0 & 0 & 0 & 0 & -(\bar{d}_t)_\times \bar{R}_t & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\bar{Q}_t = \begin{bmatrix} \mathrm{Ad}_{\bar{X}_t} & 0_{12,6} \\ 0_{6,12} & I_6 \end{bmatrix} \mathrm{Cov}(w_t) \begin{bmatrix} \mathrm{Ad}_{\bar{X}_t} & 0_{12,6} \\ 0_{6,12} & I_6 \end{bmatrix}^T$$

Reproducing the dynamics again

$$\frac{d}{dt}\begin{bmatrix} \xi_t \\ \zeta_t \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & -\bar{R}_t & 0 \\ (g)_\times & 0 & 0 & 0 & -(\bar{v}_t)_\times \bar{R}_t & -\bar{R}_t \\ 0 & I & 0 & 0 & -(\bar{p}_t)_\times \bar{R}_t & 0 \\ 0 & 0 & 0 & 0 & -(\bar{d}_t)_\times \bar{R}_t & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{A_t} \begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \\ \zeta_t^g \\ \zeta_t^a \end{bmatrix} + \underbrace{\begin{bmatrix} \mathrm{Ad}_{\bar{X}_t} & 0_{12,6} \\ 0_{6,12} & I_6 \end{bmatrix}}_{\text{noise input matrix}} \underbrace{\begin{bmatrix} w_t^g \\ w_t^a \\ 0_{3,1} \\ h_R(\tilde{\alpha}_t) w_t^v \\ w_t^{bg} \\ w_t^{ba} \end{bmatrix}}_{w_t}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & -\bar{R}_t & 0 \\ (g)_\times & 0 & 0 & 0 & -(\bar{v}_t)_\times \bar{R}_t & -\bar{R}_t \\ 0 & I & 0 & 0 & -(\bar{p}_t)_\times \bar{R}_t & 0 \\ 0 & 0 & 0 & 0 & -(\bar{d}_t)_\times \bar{R}_t & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \\ \zeta_t^g \\ \zeta_t^a \end{bmatrix} + \begin{bmatrix} \bar{R}_t & 0_{3,3} & 0 & 0 & 0 & 0 \\ (\bar{v}_t)_\times \bar{R}_t & \bar{R}_t & 0 & 0 & 0 & 0 \\ (\bar{p}_t)_\times \bar{R}_t & 0 & \bar{R}_t & 0 & 0 & 0 \\ (\bar{d}_t)_\times \bar{R}_t & 0 & 0 & \bar{R}_t & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{3\times 3} & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{3\times 3} \end{bmatrix} \begin{bmatrix} w_t^g \\ w_t^a \\ 0_{3,1} \\ h_R(\tilde{\alpha}_t) w_t^v \\ w_t^{bg} \\ w_t^{ba} \end{bmatrix}$$

$$= \begin{bmatrix} -\bar{R}_t \zeta_t^g + \bar{R}_t w_t^g \\ (g)_\times \xi_t^R - (\bar{v}_t)_\times \bar{R}_t \zeta_t^g - \bar{R}_t \zeta_t^a + (\bar{v}_t)_\times \bar{R}_t w_t^g + \bar{R}_t w_t^a \\ \xi_t^v - (\bar{p}_t)_\times \bar{R}_t \zeta_t^g + (\bar{p}_t)_\times \bar{R}_t w_t^g \\ -(\bar{d}_t)_\times \bar{R}_t \zeta_t^g + (\bar{d}_t)_\times \bar{R}_t w_t^g + \bar{R}_t h_R(\tilde{\alpha}_t) w_t^v \\ w_t^{bg} \\ w_t^{ba} \end{bmatrix}.$$

Complete Update Equations and Kalman Gain

$$(\bar{X}_t^+, \bar{\theta}_t^+) = \left(\underbrace{\exp\big(K_t^\xi \Pi \bar{X}_t Y_t\big) \bar{X}_t}_{\text{multiplicative correction}}, \underbrace{\bar{\theta}_t + K_t^\zeta \Pi \bar{X}_t Y_t}_{\text{additive correction}} \right)$$

$$P_t^+ = (I - K_t H_t) P_t (I - K_t H_t)^T + K_t \bar{N}_t K_t^T$$

$$S_t = H_t P_t H_t^T + \bar{N}_t$$

$$K_t = \begin{bmatrix} K_t^\xi \\ K_t^\zeta \end{bmatrix} = P_t H_t^T S_t^{-1}$$

$$Y_t^T = \begin{bmatrix} h_p^T(\tilde{\alpha}_t) & 0 & 1 & -1 \end{bmatrix}$$

$$H_t = \begin{bmatrix} 0 & 0 & -I & I & 0 & 0 \end{bmatrix}$$

$$\bar{N}_t = \bar{R}_t \, J_p(\tilde{\alpha}_t) \, \mathrm{Cov}(w_t^\alpha) \, J_p(\tilde{\alpha}_t)^T \, \bar{R}_t^T$$

Addition and Removal of Contact Points

Removal $\rightarrow$ from single contact to 0 contact

$$\begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix}}_{M} \begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \end{bmatrix}$$

$$\begin{cases} \xi_t^{\text{new}} = M\xi_t & \text{state update} \\ P_t^{\text{new}} = MP_tM^T & \text{covariance update} \end{cases}$$

Addition
According to forward kinematics

$$\bar{d}_t = \bar{p}_t + \bar{R}_t \cdot h_p(\tilde{\alpha}_t)$$

contact point, base attitude, and foot position relative to the base

$$\star \text{ new } d_t \text{ is inferred from } p_t$$

$$\eta_t^d = \bar{d}_t - \bar{R}_t R_t^T d_t$$

$$= \bar{p}_t + \bar{R}_t \cdot h_p(\tilde{\alpha}_t) - \bar{R}_t R_t^T d_t$$

$$= \bar{p}_t + \bar{R}_t \cdot h_p(\tilde{\alpha}_t) - \bar{R}_t R_t^T \big(p_t + R_t h_p(\tilde{\alpha}_t - w_t^\alpha)\big)$$

$$= \eta_t^p + \bar{R}_t \big(h_p(\tilde{\alpha}_t) - h_p(\tilde{\alpha}_t - w_t^\alpha)\big)$$

$$\approx \eta_t^p + \bar{R}_t J_p(\tilde{\alpha}_t) w_t^\alpha \qquad \text{first-order Taylor expansion}$$

$$\rightarrow \ \xi_t^d \approx \xi_t^p + \bar{R}_t J_p(\tilde{\alpha}_t) w_t^\alpha$$

$$\eta_t^d = \exp(\xi_t^d) \approx I + \xi_t^d \qquad (\text{only } \xi_t^R \text{ has a wedge})$$

$$\Downarrow$$

$$
\begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \\ 0 & 0 & I \end{bmatrix}}_{F_t} \begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \bar{R}_t J_p(\tilde{\alpha}_t) \end{bmatrix}}_{G_t} w_t^\alpha
$$

$$
\begin{cases} \xi_t^{\text{new}} = F_t \xi_t + G_t w_t^\alpha \\ P_t^{\text{new}} = F_t P_t F_t^T + G_t \operatorname{Cov}(w_t^\alpha)\, G_t^T \end{cases}
$$